

SPATIAL STORIES

Testing Plan for Interactive Prototype 1

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01 THREE-SENTENCE PITCH

Spatial Stories redesigns the phone-based Instagram Story workflow as a mixed-reality editing space where photos appear as floating cards around the user. Instead of navigating several 2D menus, users select and enlarge media, add and transform text or stickers, remove unwanted elements through a spatial trash zone, and publish through a clear final action. This IP1 Unity prototype simulates the overall MR workflow on desktop so the test can evaluate whether the spatial sequence and affordances are understandable before implementing full Meta Quest hand tracking.

02 TESTING OBJECTIVE

Primary unknown. The main unvalidated assumption is whether users can understand and complete the end-to-end spatial Story workflow with minimal guidance when conventional 2D menus are removed.

This test aims to discover:

- whether the yellow selection feedback and forward movement make photo selection and “grab/inspect” behaviour clear;
- whether users correctly infer the purpose of the T and S spatial tools and can directly manipulate an overlay;
- whether users understand drag-to-trash deletion and can locate the final Publish action without accidental submission;
- where hesitation, wrong actions, or facilitator prompts occur in the browse → select → edit → delete → publish sequence.

Scope note. This IP1 is intended to evaluate the conceptual workflow, discoverability and spatial organisation. It does not evaluate final Quest hand-tracking accuracy, passthrough quality, or Instagram network/backend behaviour.

03 TESTING METHODOLOGIES

Method: moderated task-based usability testing with concurrent think-aloud, structured observation, short post-task ratings, and a brief semi-structured interview. The prototype is a functional Unity horizontal/experiential prototype; mouse input simulates the intended MR pointing, pinch and direct-manipulation actions.

Participants: target 5–6 participants during the Week 5 Studio session. Participants should ideally have previous experience creating or viewing Instagram Stories; prior experience is recorded so it can be considered during analysis.

METHOD	WHY IT IS USED	EVIDENCE	LIMIT
Task-based test	Checks whether the full workflow can be completed.	Completion, time, prompts, errors.	Desktop input is a proxy for XR gestures.
Think-aloud	Reveals participant interpretation of spatial tools.	Comments, expectations, confusion.	May slightly slow task performance.
Observation	Captures hesitation and wrong actions.	First action, failed attempts, recovery.	Facilitator must avoid leading.
Post-test ratings	Gives comparable subjective measures.	1–5 ease and workflow naturalness.	Measures perceived workflow, not final hand-tracking feel.

04 PROTOTYPE DESCRIPTION / REQUIREMENTS

Prototype purpose. The Unity prototype provides enough interaction to test the complete high-level Story-creation journey rather than one isolated feature. It uses six personal travel photographs on a curved spatial album wall and guides the participant through selection, editing, removal and publishing.

PROTOTYPE ELEMENT	IMPLEMENTED IN IP1	WHAT IT ALLOWS THE TEST TO EVALUATE
Curved floating album	Six selectable photo cards with labels.	Spatial scanning and media choice.
Selection feedback	Hover response, yellow outline, selected card moves forward.	Whether selection state is visible and understandable.
Edit transition	Second click brings selected photo to the editing area; browse UI is suppressed.	Whether users understand the transition from browsing to editing.
T / S tool spheres	Text and sticker creation tools shown in a fixed spatial zone.	Tool discoverability and correct tool choice.
Overlay manipulation	Drag to move; scroll to scale; Z/X to rotate in the desktop simulation.	Direct-manipulation concept and control understanding.
Trash zone	Overlay can be dragged to the left-side trash zone and released to delete.	Deletion affordance and error recovery.
Publish feedback	Publish appears after an edit, gives press feedback, moves the photo, then confirms success.	Completion confidence and accidental submission risk.
Reset + logs	Reset prepares the next session; event logging records interaction steps.	Repeatable testing and quantitative evidence.

05 CONSTRUCTION DECISIONS: INCLUDED AND EXCLUDED

Included because they are necessary for the testing objective:

- the complete browse → select → edit → delete → publish journey;
- strong visible state changes (yellow selection, dimmed browse state, tool appearance, trash feedback, publish confirmation);
- direct manipulation of overlays and a spatial delete action;
- real photographs rather than abstract placeholders, so participants experience a believable Story context;
- reset and event logging so multiple in-class sessions can be run consistently.

Deliberately excluded from IP1:

- Meta Quest hand tracking and MR passthrough - these are reserved for later XR prototypes; IP1 isolates the high-level workflow;
- Instagram login, social graph, uploading and network behaviour - these do not affect the current interaction assumptions;
- full text-entry keyboard and large sticker catalogue - preset content is sufficient to test tool discovery and spatial placement;
- production-level visual effects and accessibility settings - polish is limited to what is needed for a clear, stable test.

06 DATA COLLECTION METHOD

During each test, the facilitator records both behavioural and subjective data. A participant ID (P01, P02, etc.) is used instead of names. The Unity interaction log can support the observation notes, but the facilitator sheet remains the primary record because it captures hesitation, verbal comments and reasons for failure.

METRIC	RECORD	PURPOSE
Overall task completion	Complete / Partial / Failed	Whether the end-to-end workflow is usable.
Time to completion	Seconds	Efficiency and unusually long hesitation.
Prompts	0 / 1 / 2+	How much facilitator help was required.
First correct selection action	Yes / No + seconds	Discoverability of photo selection.
Correct T/S tool choice	Yes / No	Meaning of the editing tools.
Overlay manipulation	Move / scale / rotate success	Understanding of direct manipulation.
Delete success	Yes / No + hesitation	Whether drag-to-trash is understood.
Publish success	Yes / No + accidental actions	Whether the final action is clear.
Ease rating	1–5	Perceived ease of the workflow.
Workflow naturalness	1–5	How natural the spatial sequence feels conceptually.
Critical comment	Short quote / paraphrase	Actionable qualitative feedback.

07 TESTING SETUP

1. Open the correct Unity project in Unity 6000.3.20f1 and open the SpatialStories_IP1 scene.
2. Connect a mouse/trackpad and confirm the Game view is visible at a comfortable size.
3. Run one full facilitator check: select photo → enter edit mode → add overlay → move/scale/rotate → delete → publish → reset.
4. Confirm Console has no red errors and the participant can see the on-screen controls.
5. Prepare a timer and one copy of the data-recording sheet for each participant.
6. Start from a fresh reset state before every participant. Do not leave an edited Story from the previous test.
7. Seat the participant directly in front of the display and explain only the desktop-to-MR input mapping; do not explain the meaning of T, S, Trash or Publish before the task.
8. If a participant is stuck for 10 seconds or explicitly says they cannot continue, give the standardised prompt listed below and record one prompt.

08 FACILITATOR INTRODUCTION AND TASK

Read this introduction consistently:

“This prototype explores creating an Instagram Story in a mixed-reality space. The desktop controls simulate future MR gestures, so please focus on whether the spatial workflow makes sense rather than judging mouse control as the final interaction. Please think aloud and say what you expect each object to do. I will mostly observe and will only help if you cannot continue.”

Explain only these control mappings:

- Left click = select / simulated pinch.
- Left-click drag = grab / move an editable object.
- Mouse wheel or trackpad scroll = scale a selected overlay.
- Z / X = rotate a selected overlay.
- Right-drag = look; WASD = move if needed.
- R = reset (facilitator use if required).

Participant task:

“Imagine you want to create and publish an Instagram Story. Choose one photo, bring it into the editing space, add the text and sticker tools, reposition at least one item, remove one unwanted item, and publish the finished Story. Please complete the task in the way that feels most natural to you.”

09 TESTING PROCESS AND TIMING (TARGET: 5 MINUTES)

TIME	FACILITATOR ACTION	PARTICIPANT ACTION	RECORD
0:00–0:30	Give introduction and control mapping. Start timer when task begins.	Listen; ask clarification only about controls.	Prior Instagram Story familiarity.
0:30–1:15	Observe without coaching.	Browse and select a photo; enter edit mode.	Time to first correct action; wrong selections; prompt count.
1:15–2:45	Observe tool discovery and manipulation.	Use T and S; move one overlay; scale or rotate if understood.	Correct tool choice; manipulation success; think-aloud comments.
2:45–3:30	Observe deletion. Do not point at Trash.	Remove one unwanted overlay.	Delete success; hesitation; accidental action.
3:30–4:00	Observe final action.	Publish Story.	Publish success; accidental submission; completion time.
4:00–5:00	Ask ratings and 3 short questions.	Rate and explain experience.	Ease, naturalness, most confusing part, suggested change.

Standardised help rule: after 10 seconds with no progress, say only “Look around the spatial interface for an object that might help with your current goal.” If still stuck after another 10 seconds, give a task-specific hint (for example, “Try the object on the left”). Record every prompt. Do not demonstrate the action unless the test must be recovered.

10 POST-TEST QUESTIONS

9. On a scale from 1 (very difficult) to 5 (very easy), how easy was the overall workflow?
10. On a scale from 1 (not natural) to 5 (very natural), how natural did the spatial sequence feel as a concept?
11. Which action was the most confusing or required the most thought?
12. Was there any object or feedback that did not behave as you expected?
13. If you could change one thing before this becomes a real MR/hand-tracking version, what would you change?

11 SUCCESS CRITERIA AND INTERPRETATION

Primary success indicator. A core interaction is considered promising when at least 80% of participants complete it without facilitator help, there is no repeated critical error, and the relevant ease/naturalness rating averages 4/5 or higher. This continues the threshold defined in the initial concept testing plan.

RESULT	INTERPRETATION	NEXT DESIGN ACTION
≥80% success, 0 prompts, average ≥4/5	Interaction is sufficiently clear for the current stage.	Keep the interaction and test it with real XR input in Prototype 2.
60–79% success or repeated hesitation	Affordance is partially understood but not reliable.	Refine feedback, placement, label or timing before the next prototype.
<60% success or repeated critical error	The interaction model is not sufficiently discoverable.	Rework the interaction, not just its visual polish, then retest.
High completion but low naturalness	Users can operate it but the spatial metaphor may feel forced.	Explore a different spatial action or reduce unnecessary steps.

12 PARTICIPANT DATA-RECORDING SHEET

ID	Story exp.	Complete	Time	Prompts	T/S	Delete	Publish	Ease / Nat.
P01		C / P / F		0 / 1 / 2+	Y / N	Y / N	Y / N	__/5 __/5
P02		C / P / F		0 / 1 / 2+	Y / N	Y / N	Y / N	__/5 __/5
P03		C / P / F		0 / 1 / 2+	Y / N	Y / N	Y / N	__/5 __/5
P04		C / P / F		0 / 1 / 2+	Y / N	Y / N	Y / N	__/5 __/5
P05		C / P / F		0 / 1 / 2+	Y / N	Y / N	Y / N	__/5 __/5
P06		C / P / F		0 / 1 / 2+	Y / N	Y / N	Y / N	__/5 __/5

Observation notes (one short line per participant):

P01: _____

P02: _____

P03: _____

P04: _____

P05: _____

P06: _____

13 DESIGN JOURNAL / PROTOTYPE DEVELOPMENT SUMMARY

The following summary documents the design decisions that led to this IP1. It can be supported in GitHub with screenshots, commits and short captions showing the actual Unity changes.

ITERATION	WHAT CHANGED	WHY
Week 2 - low-fidelity concept	Paper prototype explored a floating album, highlighted media, T/S tools, left-side bin and Submit action.	Established the complete spatial Story workflow.
Concept refinement	Added yellow selection feedback, enlargement for inspection, separate T/S spheres, fixed trash/publish zones.	Improved discoverability and sequence.
Unity horizontal prototype	Implemented an end-to-end desktop simulation of the MR workflow.	Made the concept testable without waiting for full XR hardware implementation.
Personal media + spatial album	Replaced placeholder images with six personal travel photos and refined the wall layout.	Made the scenario believable and the album visually varied.
Selection refinement	Added hover scale, stronger yellow outline, forward movement and brighter selected caption.	Strengthened spatial affordance for selection.
Edit-mode refinement	Selected photo moves forward; browse UI and labels are hidden; tools appear after the transition.	Separates browsing and editing into clear states.
Direct manipulation	Added drag, scale and rotate for overlays.	Tests whether editing can feel object-based rather than menu-based.
Drag-to-trash	Changed deletion from click-delete to spatial drag-and-release.	Aligns deletion with the embodied MR concept.
Publish refinement	Added publish-button feedback, photo movement and delayed completion confirmation.	Makes the final action readable and testable.

14 WEEK 5 DELIVERABLES CHECKLIST

- ☐ Testing plan complete before class.
- ☐ Functional Unity 3D prototype with programming and a stable test path.
- ☐ Source code / Unity project prepared for submission.
- ☐ Short video of the working prototype prepared.
- ☐ Minimum target of 5 in-class participant tests, with actionable data recorded.
- ☐ Testing results summarised and posted to GitHub at the end of class.
- ☐ Statement of Originality and AI/source acknowledgement included in Git.

15 REFERENCES / SOURCE BASIS

- DECO2300/7230 Digital Prototyping and Extended Reality. (2026). Interactive Prototype 1 - Unity Horizontal prototype brief and testing plan template.
- University of Queensland. (2026). DECO7230 course profile - Interactive Prototype 1 assessment requirements.
- Yang, Z. (2026). Spatial Stories: Redesigning Instagram Story Creation for Mixed Reality - Concept Design Report.

APPENDIX A STATEMENT OF ORIGINALITY AND AI ACKNOWLEDGEMENT DRAFT

Statement of originality. I certify that this submission is my own work except where the use of external sources and generative AI tools is acknowledged below. I reviewed, tested and adapted the prototype, code, testing decisions and written material before submission, and I am able to explain the purpose and operation of the submitted work.

AI acknowledgement table (for GitHub). The exact wording can be transferred to the course/UQ acknowledgement format used for submission.

TOOL	USE	PROMPT (ENGLISH)	SECTION / ARTEFACT	DATE
ChatGPT	Suggested a modular Unity interaction structure and helped refine the spatial selection, editing, trash and publish behaviours. I implemented/tested the project in Unity and reviewed each iteration.	Suggest a practical Unity structure for my Spatial Stories horizontal prototype, focusing on a floating photo wall, selection feedback, text/sticker editing, trash and publishing.	Unity prototype structure and interaction iterations	16 Aug 2026
ChatGPT	Helped diagnose Unity compilation/runtime issues from screenshots and suggested targeted code fixes. I verified the fixes in Unity before continuing.	Help me identify the cause of this Unity Console error and suggest the smallest fix that preserves the current prototype.	Prototype debugging and code refinement	16 Aug 2026
ChatGPT	Helped draft and organise the IP1 testing plan from my existing concept, prototype and assignment brief. I reviewed the tasks, metrics and success criteria for alignment with what I will test in class.	Using my Spatial Stories concept, current Unity prototype and the DECO7230 IP1 brief, help me draft a robust five-minute testing plan with objective, methodology, metrics, setup and participant procedure.	IP1 testing plan	16 Aug 2026